Master of Business Analytics

Prescriptive Analytics & Optimization

MBAN 5540

Week 09: Monte Carlo Simulation

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What is Simulation?

- An attempt to duplicate the features, appearance, and characteristics of a real system

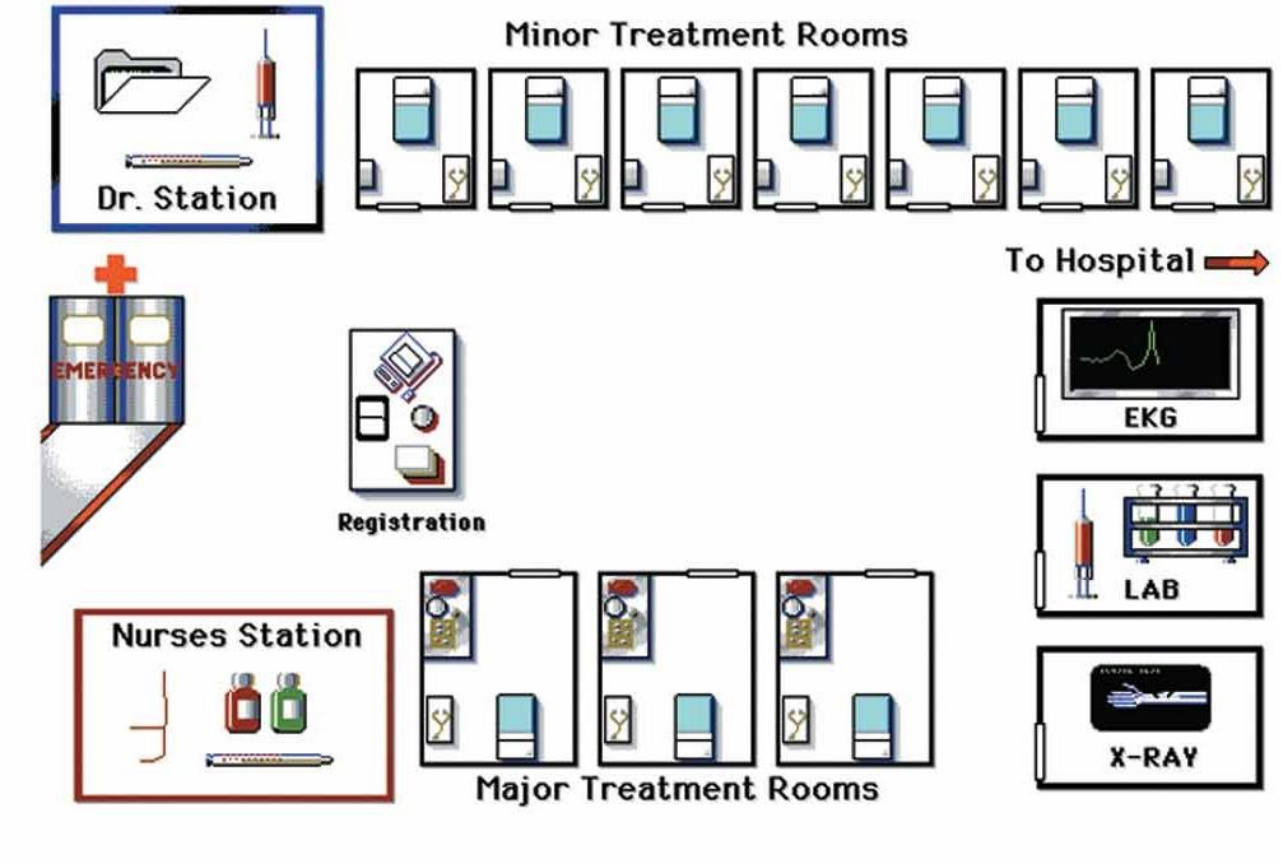
  1. To imitate a real-world situation mathematically
  2. To study its properties and operating characteristics
  3. To draw conclusions and make action decisions based on the results of the simulation

- Simulation is used for complex systems, when no analytical or closed-form solution is available.

Simulation Applications

Ambulance location and dispatching	Bus scheduling
Assembly-line balancing	Design of library operations
Parking lot and harbour design	Taxi, truck, and railroad dispatching
Distribution system design	Production facility scheduling
Scheduling aircraft	Plant layout
Labour-hiring decisions	Capital investments
Personnel scheduling	Production scheduling
Traffic-light timing	Sales forecasting
Voting pattern prediction	Inventory planning and control

Role of Computer Analysis



Simulation Steps

Define	Define the problem
Introduce	Introduce the important variables associated with the problem
Construct	Construct a numerical model
Set up	Set up possible courses of action for testing by specifying values of variables
Run	Run the experiment
Consider	Consider the results (possibly modifying the model or changing data inputs)
Decide	Decide what course of action to take

Advantages of Simulation

- Relatively straightforward and flexible
- Can be used to analyze large and complex real-world situations that cannot be solved by conventional models
- Real-world complications can be included that most models cannot permit
- "Time compression" is possible
- Allows "what-if" types of questions
- Does not interfere with real-world systems
- Can study the interactive effects of individual components or variables in order to determine which ones are important

Disadvantages of Simulation

- Can be very expensive and may take months to develop
- It is a trial-and-error approach that may produce different solutions in repeated runs
- Managers must generate all the conditions and constraints for solutions they want to examine
- Each simulation model is unique

Monte Carlo Simulation

The Monte Carlo method may be used when the model contains elements that exhibit chance in their behavior.

1. Set up probability distributions for important variables
2. Build a cumulative probability distribution for each variable
3. Establish an interval of random numbers for each variable
4. Generate random numbers
5. Simulate a series of trials

Probability of Demand

(1) Demand for Tires	(2) Frequency	(3) Probability of Occurrence	(4) Cumulative Probability
0	10	10/200 = .05	.05
1	20	20/200 = .10	.15
2	40	40/200 = .20	.35
3	60	60/200 = .30	.65
4	40	40/200 = .20	.85
5	30	30/ 200 = .15	1.00
	200 days	200/200 = 1.00	

Assignment of Random Numbers

Daily Demand	Probability	Cumulative Probability	Interval of Random Numbers
0	.05	.05	0 through .05
1	.10	.15	.06 through .15
2	.20	.35	.16 through .35
3	.30	.65	.36 through .65
4	.20	.85	.66 through .85
5	.15	1.00	.86 through 1

Simulation Example

Day Number	Random Number	Simulated Daily Demand
1	.52	3
2	.37	3
3	.82	4
4	.69	4
5	.98	5
6	.96	5
7	.33	2
8	.50	3
9	.88	5
10	.90	5
Average		**3.9**

When to Stop?

In the previous example, we run the model for 10 days, and get the average of 3.9.

- How good is this number?
- Will it change if we run the model for 100 days instead?
- When is the best time to finish a model run?

Steady-state

- If a system is in a steady state, then the recently observed behavior of the system will continue into the future.

- In simulation, we cannot draw any conclusion unless the simulation system is in steady-state.

- It might be expensive(time, cost, etc.) to reach to a steady-state! We need to find the balance between:
  - The cost of reaching to a steady-state
  - The cost of having not the best results out of the system

Using Software in Simulation

- Computers are critical in simulating complex tasks

- General-purpose languages - PYTHON, C++

- Special-purpose simulation languages - GPSS, SIMSCRIPT
  1. Require less programming time for large simulations
  2. Usually more efficient and easier to check for errors
  3. Random-number generators are built in

Using Software in Simulation

- Commercial simulation programs are available for many applications:
  - Extend
  - Modsim
  - Enterprise Dynamics
  - Simfactory
  - ProModel
  - Micro Saint
  - ARENA

- Spreadsheets such as Excel can be used to develop some simulations

Example 1

- Two fair coins are tossed at the same time. If both show heads or tails, we get $10, otherwise we get nothing.

- Simulate this process for 50 tosses.
  - How much do we expect to earn, on average, per toss?
  - What is the expected earn when the system reaches a steady-state?

  - Please check the Excel file: Simulation Examples in Lecture.

Example 2

- There are 3 closed doors in the Monty Hall show. One of the doors hides a prize and the other two doors have no prize.
- You state out which door you pick, but you do not open it right away. The host opens one of the other two doors, and there is no prize behind it. Now, there are two closed doors, one of which you picked initially. The prize is behind one of the closed doors, but you do not know which one.
- The host asks you "Do you want to switch doors?"
- What do you do? Is there a difference between staying with your initial door, or switch to the other door?

Use simulation to answer!

Example 3: Newsvendor Problem

- Cost of understocking per unit: c_u = Sales price per unit – Cost per unit = $p - c$
- Cost of overstocking per unit : c_o = Cost per unit – salvage value = $c - s$
- Total cost: $c_o\, Max(0, Q - D) + c_u\, Max(0, D - Q) + cQ$
- Total Revenue: $p.\min(D, Q)$
- What is the expected profit if:

$$p = 25$$
$$c = 15$$
$$s = 5$$

Use simulation to answer.

Example 4: A Queuing Model

- A bank with one server collected data for arrivals and services in the past and generated the below probability distributions:

- Use simulation to find:
  - The average waiting time
  - The average time in the system
  - The average length of queue

Arrival Distribution	
Probability	Time Between Arrivals (minutes)
0.10	0
0.15	1
0.10	2
0.35	3
0.25	4
0.05	5

Service Distribution	
Probability	Service time (minutes)
0.25	1
0.20	2
0.40	3
0.15	4